

Body-fitted finite-volume DFG 2D-1 cylinder verification

Actual body-fitted Gmsh meshes; strict physical relative error <3%.

Scheme: {'momentum_convection': 'upwind', 'pressure_gradient': 'least-squares', 'face_interpolation': 'arithmetic', 'wall_gradient': 'displacement'}

CPU calculations; no new GPU or matched TensorLBM result.

Positive mesh geometry and nonorthogonality gate precede all solves.

upwind: 8252 cells; converged=True; accepted=False

{'drag': 5.797085008150745, 'lift': 0.04843301350411677, 'pressure_difference_pa': 0.11582988221321745}

{'drag_relative_error': 0.0389907, 'lift_relative_error': 3.5609992, 'pressure_difference_pa_relative_error': 0.0143829}

linear-upwind: 8252 cells; converged=True; accepted=False

{'drag': 5.516257331593865, 'lift': 0.023351854915401, 'pressure_difference_pa': 0.11381505757238523}

{'drag_relative_error': 0.0113411, 'lift_relative_error': 1.1990742, 'pressure_difference_pa_relative_error': 0.0315274}

conservative-linear-upwind: 8252 cells; converged=True; accepted=False

{'drag': 5.543013464759672, 'lift': 0.010060106247246643, 'pressure_difference_pa': 0.11517945530273684}

{'drag_relative_error': 0.0065457, 'lift_relative_error': 0.0526269, 'pressure_difference_pa_relative_error': 0.0199175}

Full independently replayed equations: True

Refinement sequence accepted: False

Asymptotic grid regime qualified: False

Problem, numerical method and reproducibility

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Problem and reference

DFG 2D-1: steady incompressible laminar flow, $Re=20$, channel 2.2 by 0.41 m, cylinder center (0.2,0.2) m and radius 0.05 m. The mean inlet speed is 0.2 m/s, density 1 kg/m³ and dynamic viscosity 0.001 Pa s. No-slip applies to the cylinder and channel walls. The inlet parabolic velocity is integrated exactly over each physical face; outlet pressure is zero with zero-gradient velocity. Drag, lift and front/rear pressure difference are compared with the [FeatFlow reference](https://wwwold.mathematik.tu-dortmund.de/~featflow/en/benchmarks/cfdbenchmarking/flow/dfg_benchmark1_re20.html).

Mesh and discretization

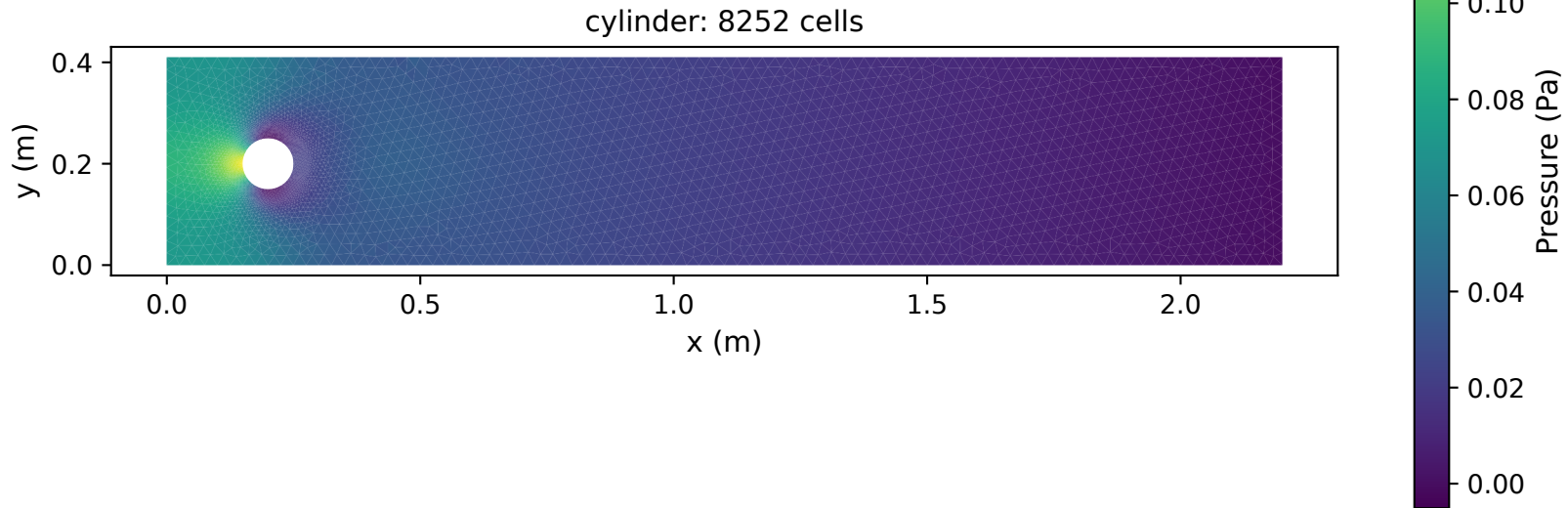
Gmsh 4.15.2 frontal Delaunay generates conforming triangles with physical boundary tags; cylinder CAD arcs or a closed NACA0012 interpolating spline define the body. Physical wall faces are linear chords. See the [Gmsh official manual](<https://gmsh.info/doc/texinfo/gmsh.html>) for algorithm 6 and Distance/Threshold fields. Neither Cartesian obstacle masks nor staircase boundaries are used. Positive volumes and projected face distances and a maximum nonorthogonality below 70 degrees are required before solving. The 70-degree project gate is a screening criterion, not proof of sufficient accuracy.

The recorded discretization is `{"face_interpolation": "arithmetic", "momentum_convection": "upwind", "pressure_gradient": "least-squares", "wall_gradient": "displacement-correction"}`. Linear upwind uses the least-squares gradient of the upstream cell and the actual face-to-cell displacement; its conservative deferred correction is inspired by [OpenFOAM linearUpwind](<https://github.com/OpenFOAM/OpenFOAM-dev/blob/master/src/finiteVolume/interpolation/surfaceInterpolation/schemes/linearUpwind/linearUpwind.C>). This is unlimited reconstruction, so no bounded/TVD claim is made. Gauss pressure gradients use geometric face interpolation and the same boundary face pressure used by pressure traction. Nonorthogonal diffusion, physical Rhie-Chow flux and no-slip reconstructed molecular stress are retained. Cylinder acceleration uses a full-grid coupled Picard matrix with an independent response check; NACA uses SIMPLE and Anderson mixing. CPU sparse solves check true residuals. No reference value enters the equations, initialization or relaxation.

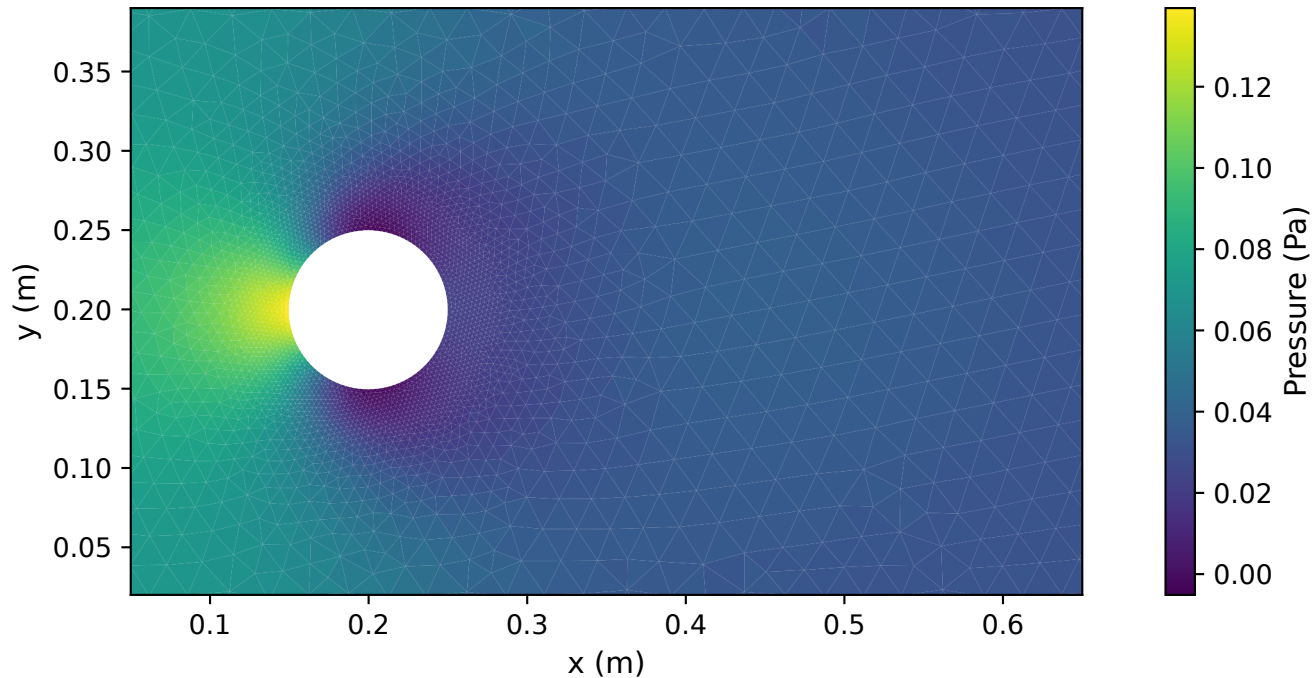
Reproduction and acceptance

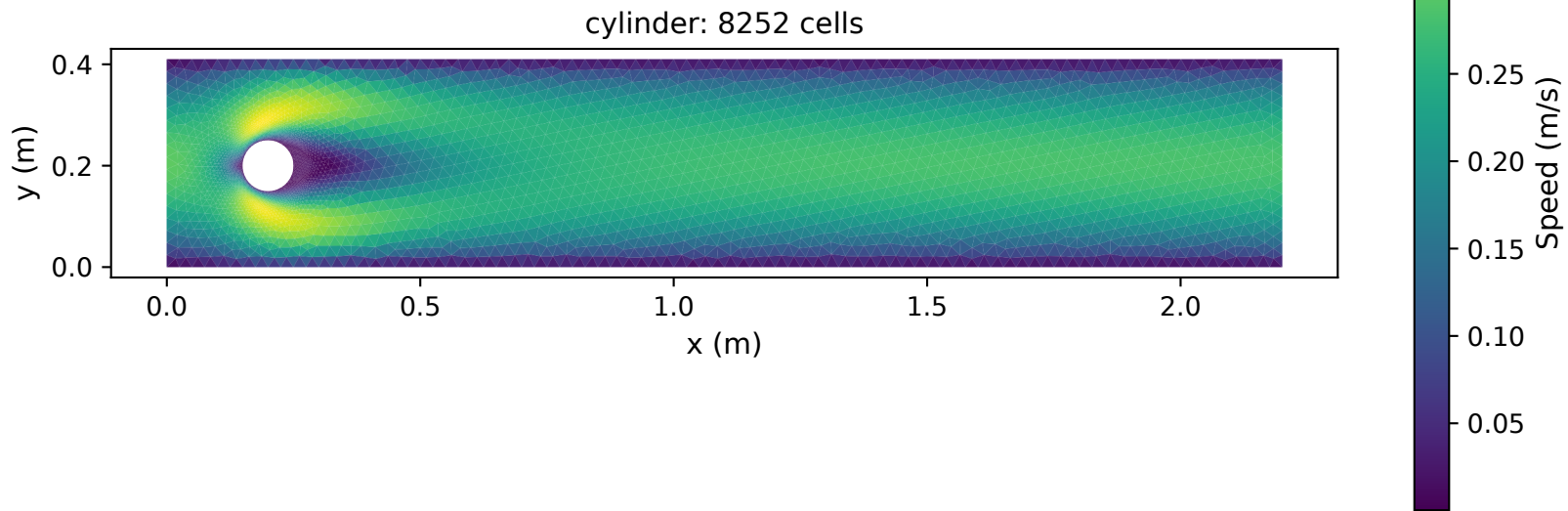
```
```bash
uv pip install --python /path/to/python gmsh numpy scipy matplotlib
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 PYTHONPATH=src python -m
tensorfvm.verification.external_gmsh --case cylinder --scales 1.0 1.0 1.0 --solver upwind --max-
iterations 300 --output results/cylinder
PYTHONPATH=src python scripts/audit_external_equations.py results/cylinder
PYTHONPATH=src python -m tensorfvm.verification.external_mesh_report results/cylinder
```
```

Raw mesh/field files, velocity/pressure arrays, face mass fluxes, separate tractions, full histories, linear residuals and SHA256 source/artifact manifests accompany this report. Each physical relative error must be strictly below 3%; steady equation residuals must be below 1e-6 and the physical flux fixed-point defect below 1e-7. Source-preserving independent NumPy replay checks full configured equations. Three runs do not establish an asymptotic grid regime; qualification of the finest run does not qualify the entire refinement sequence. GPU and matched TensorLBM runs have not been performed for this new unstructured path.

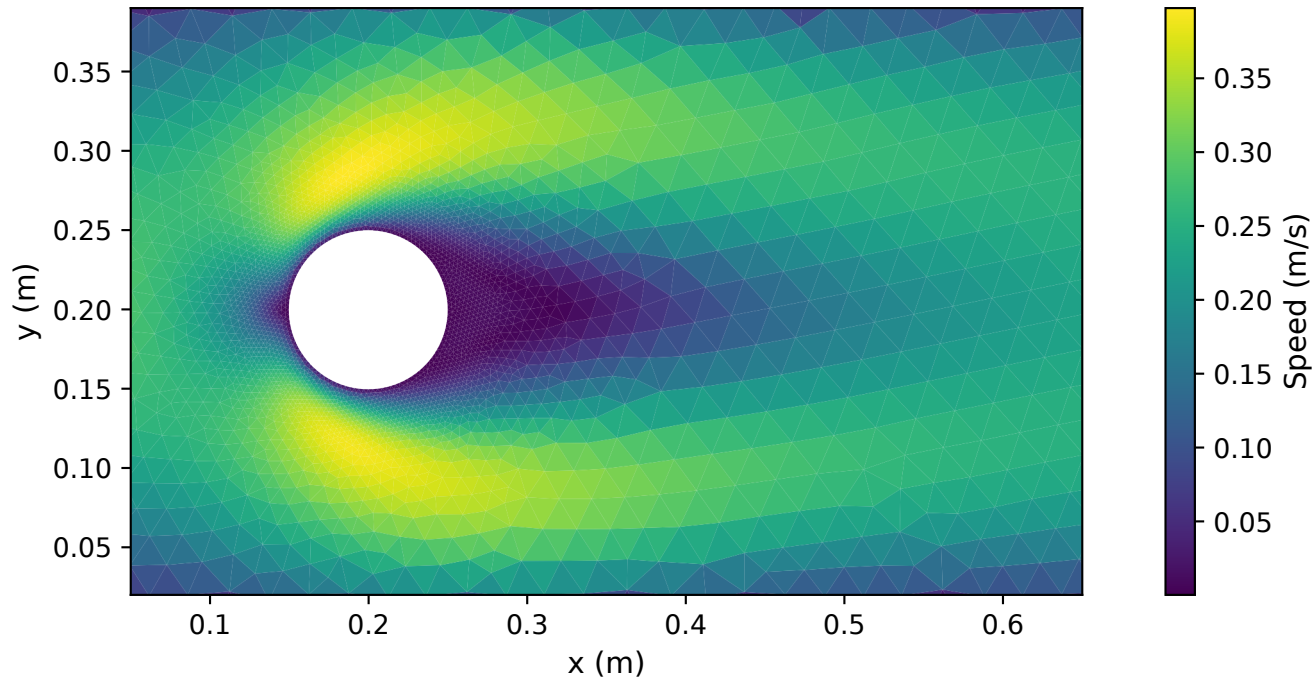


cylinder: 8252 cells

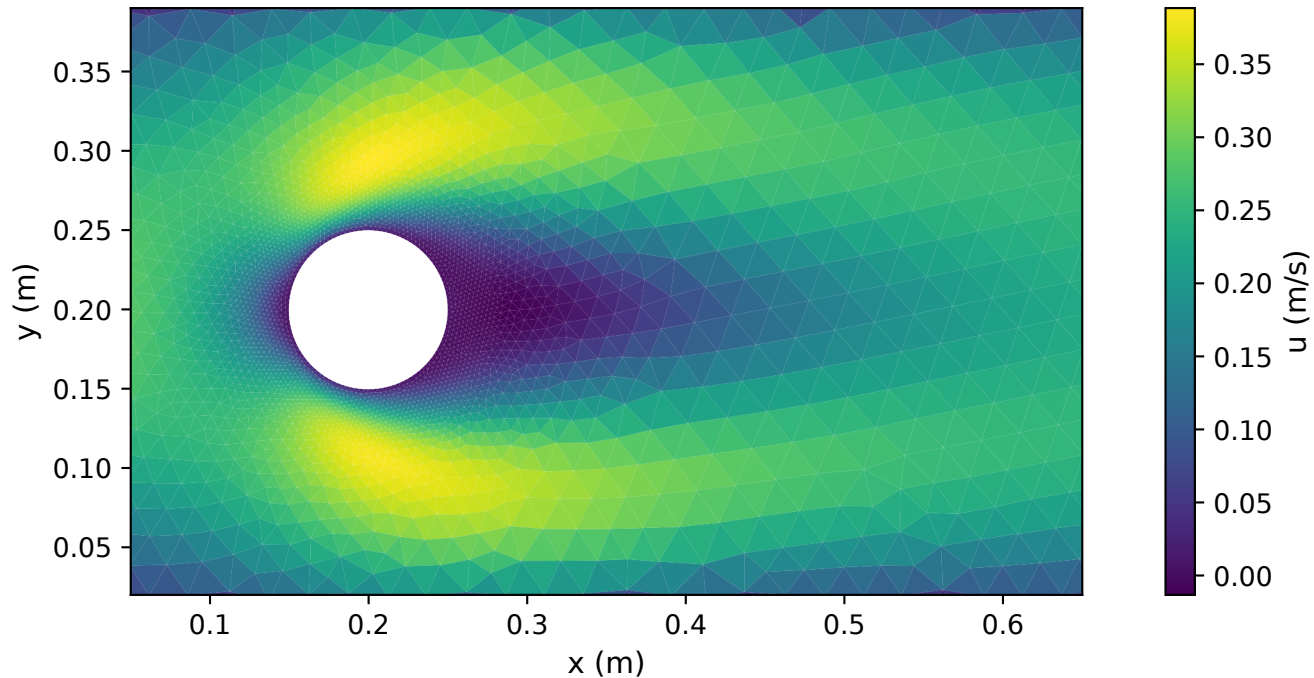




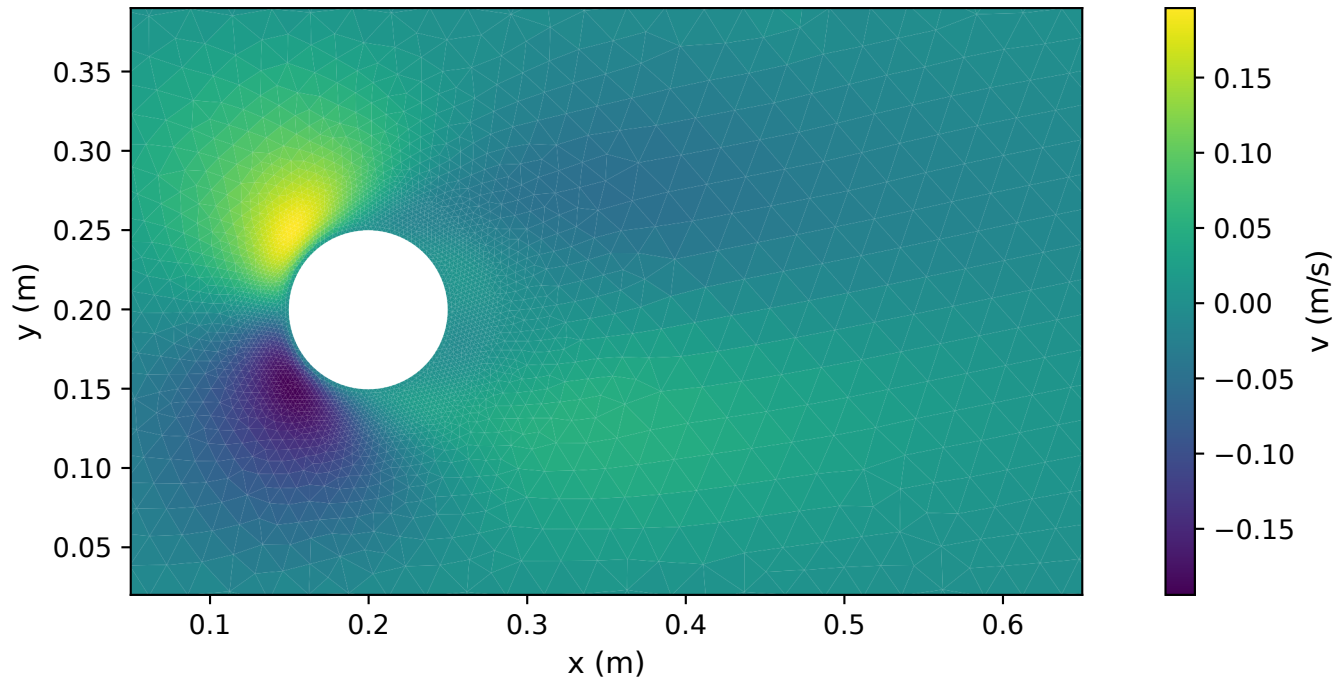
cylinder: 8252 cells



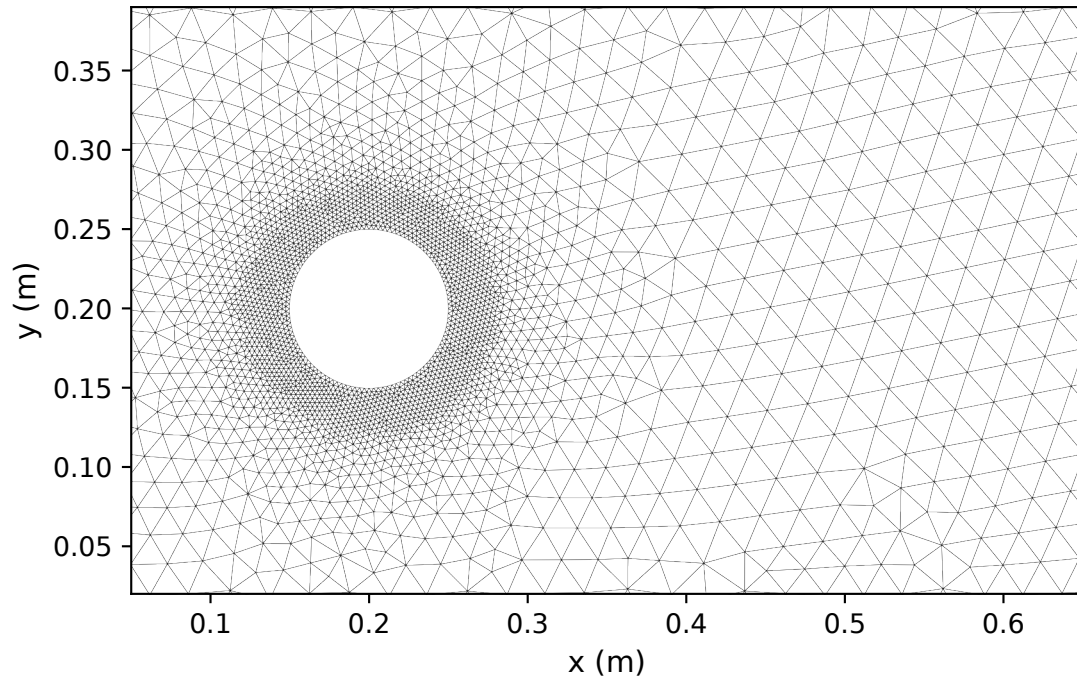
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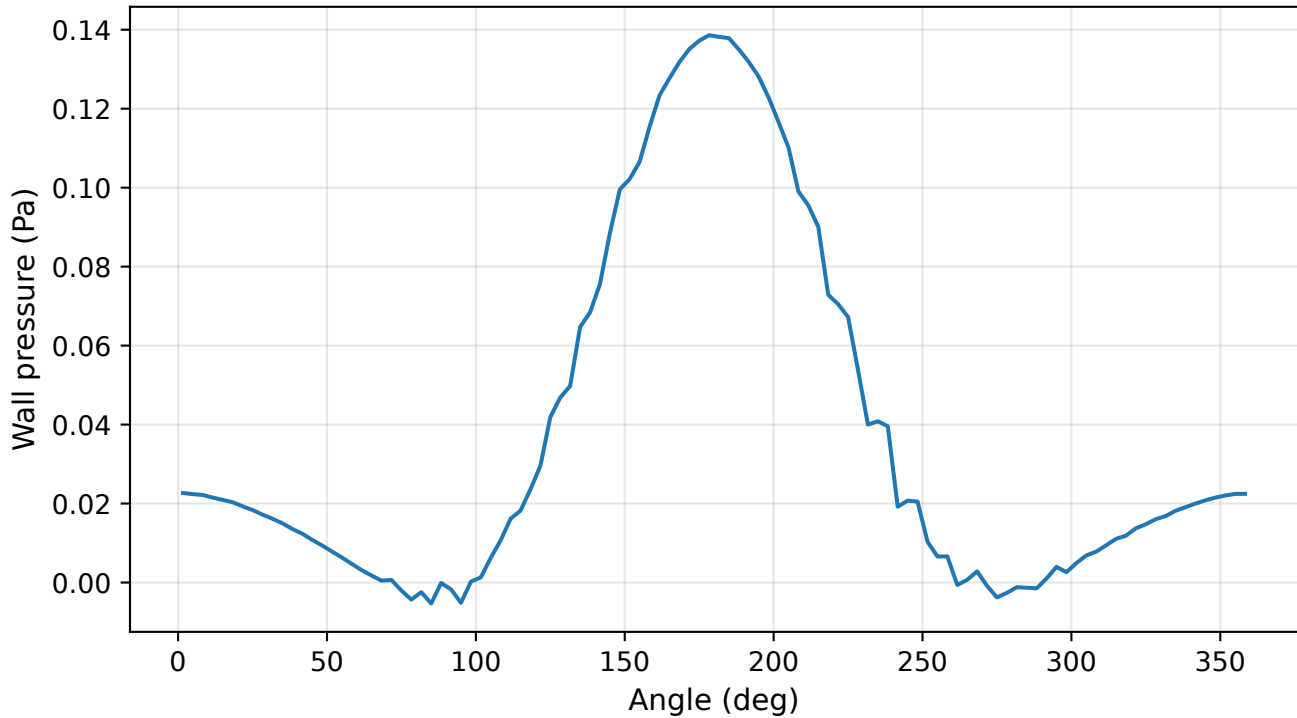


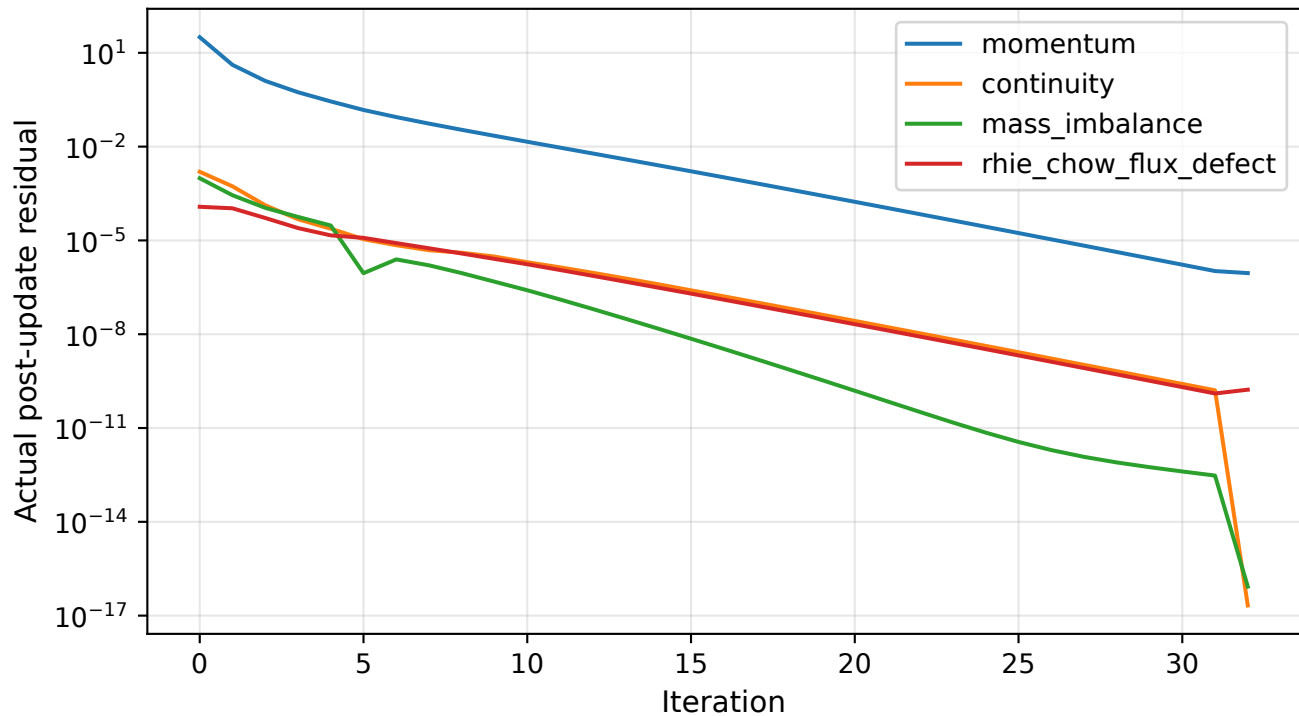
cylinder: 8252 cells

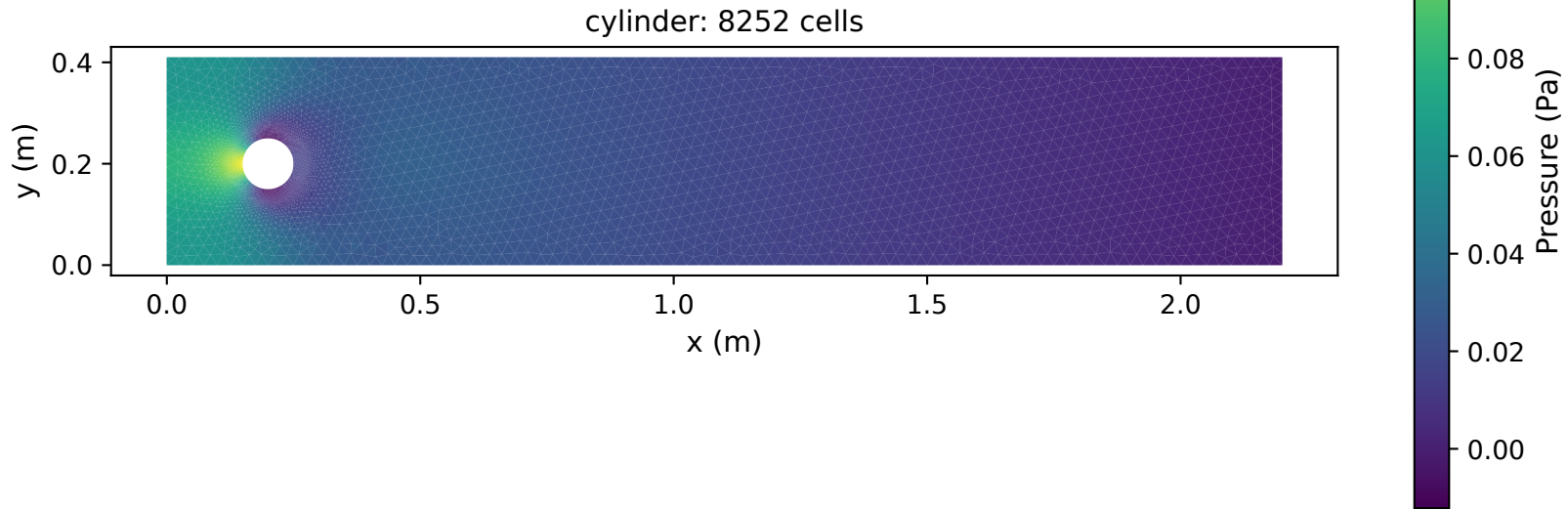


Actual body-fitted mesh

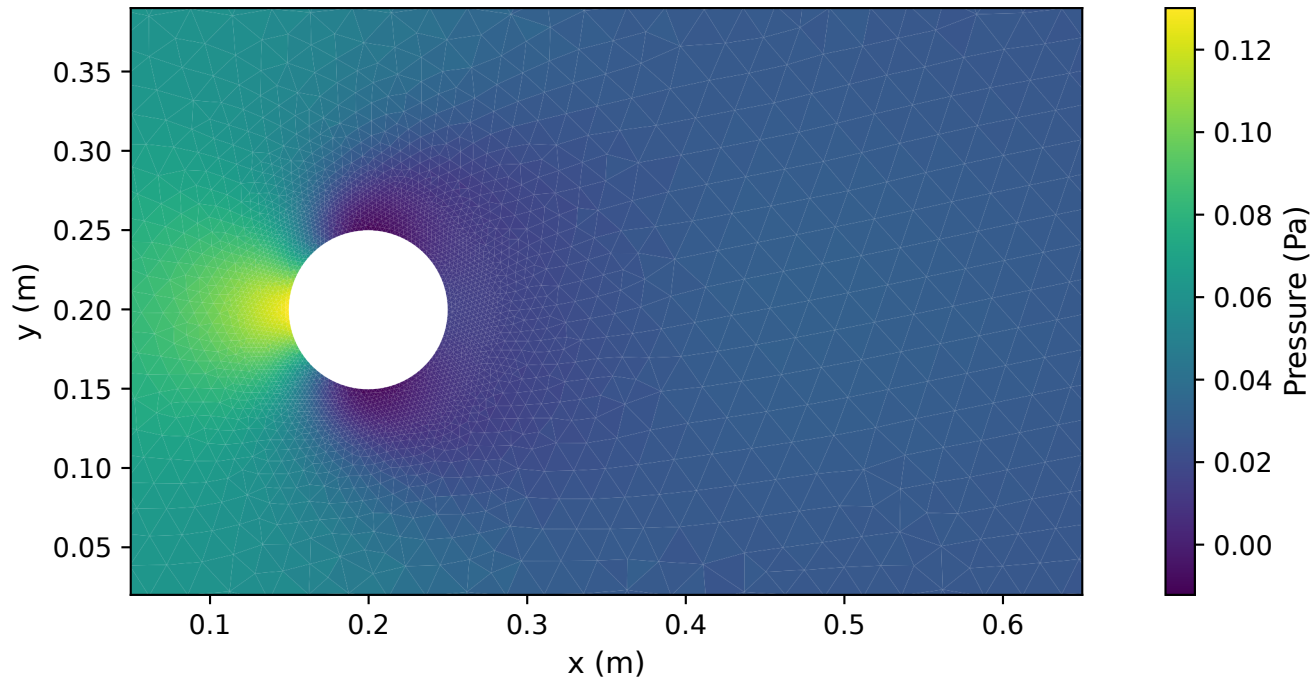


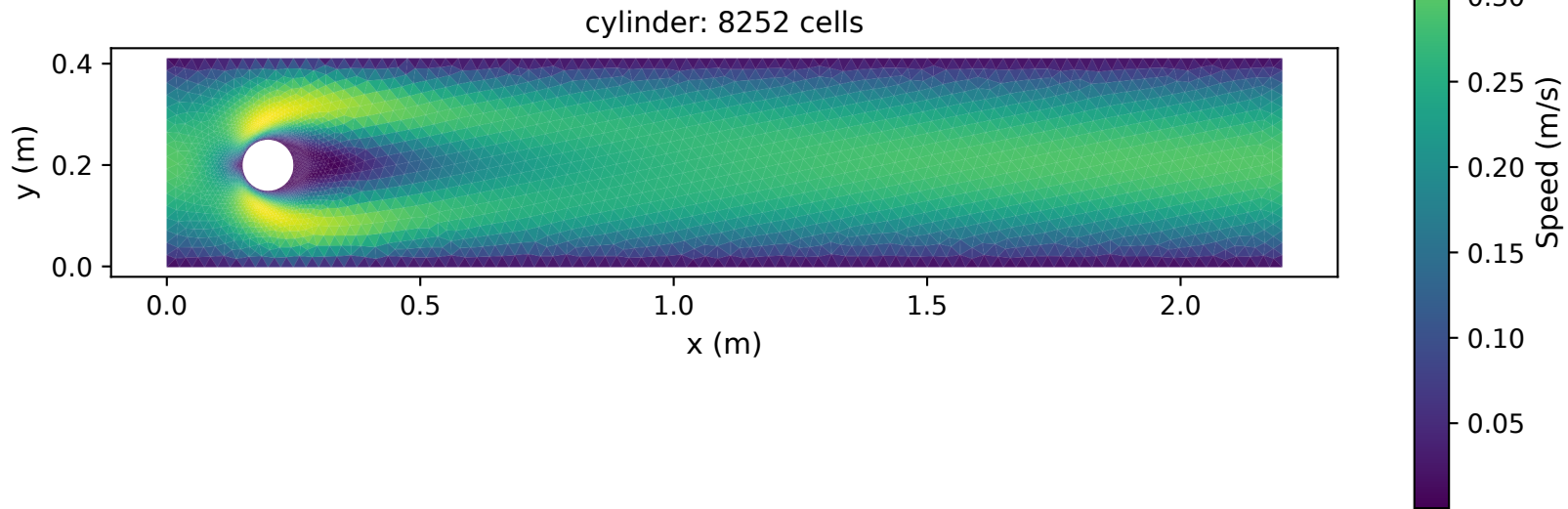




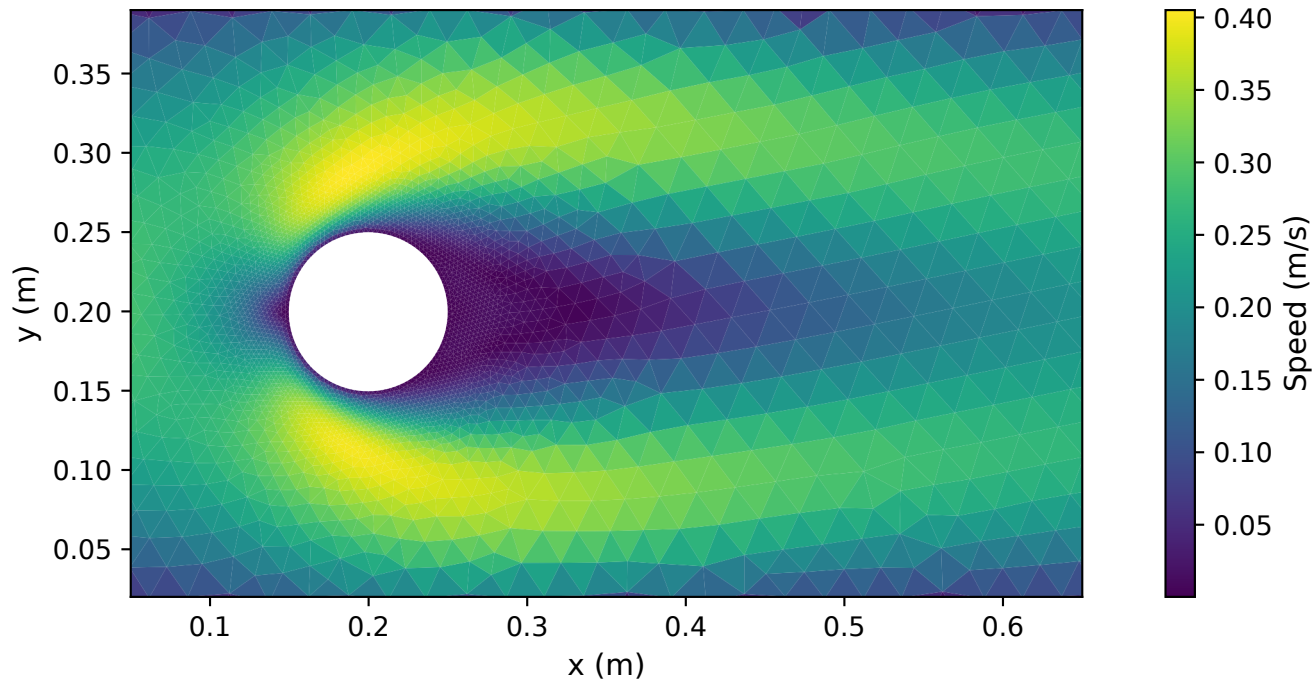


cylinder: 8252 cells

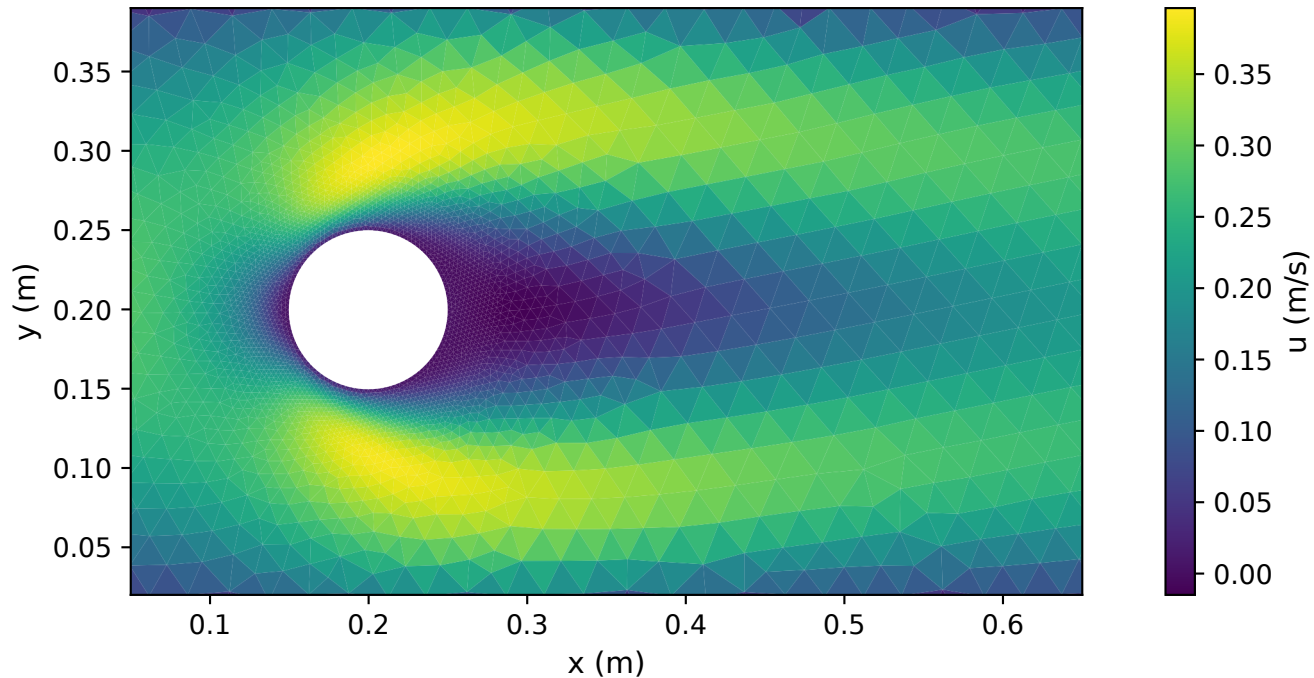




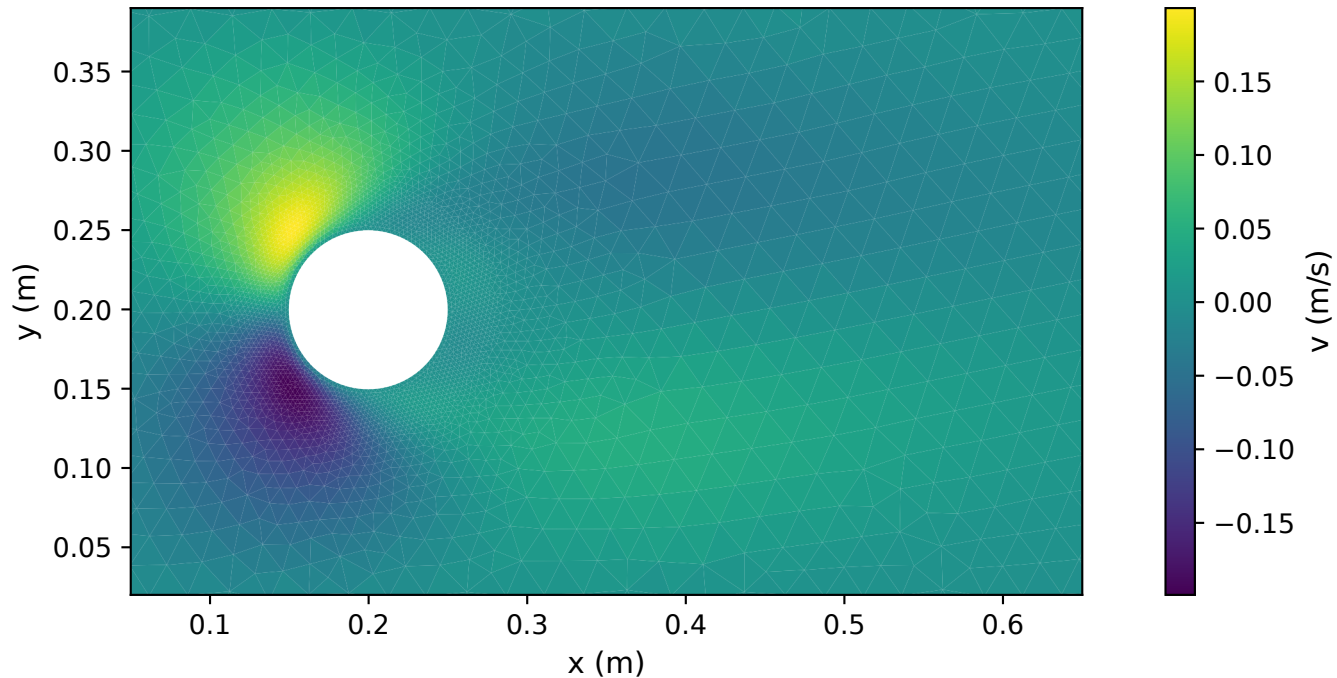
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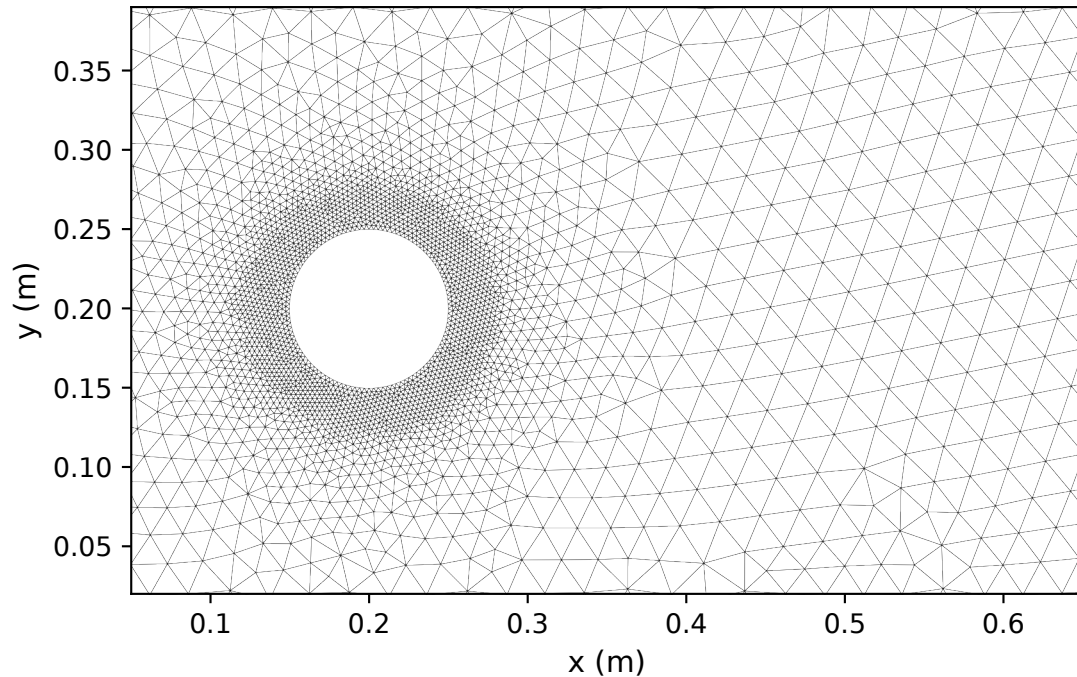
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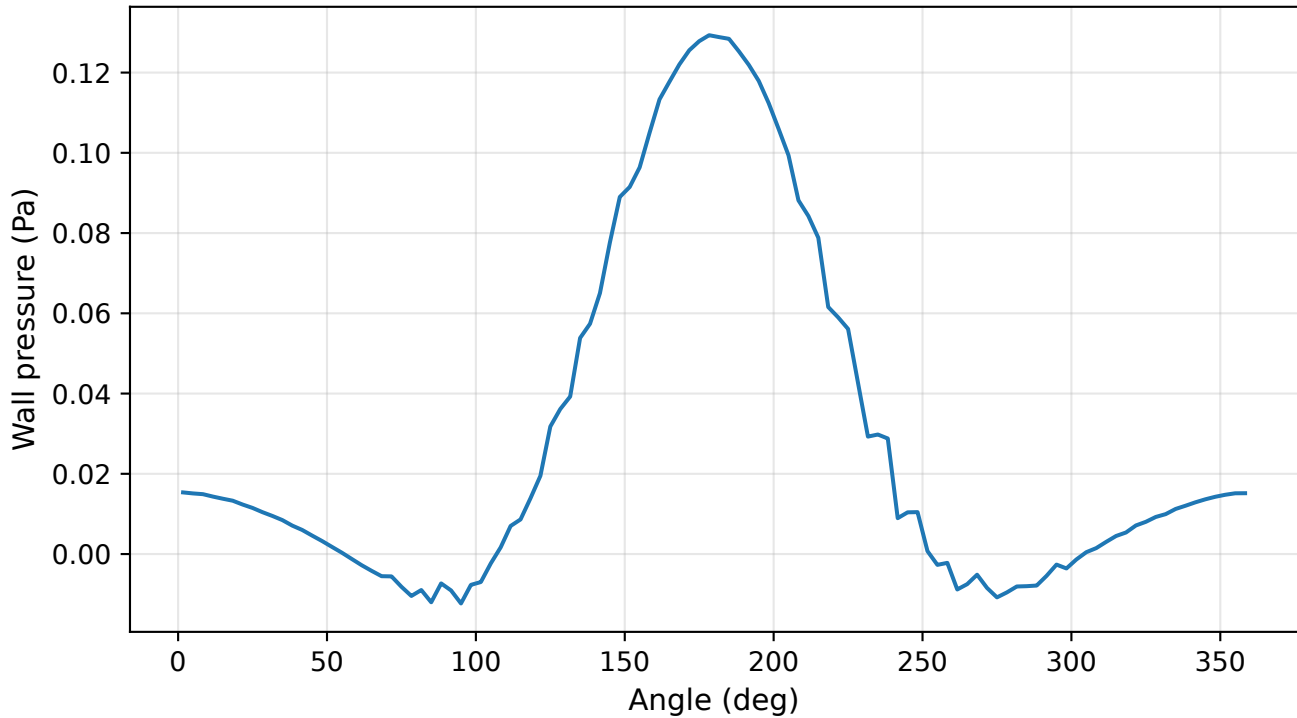


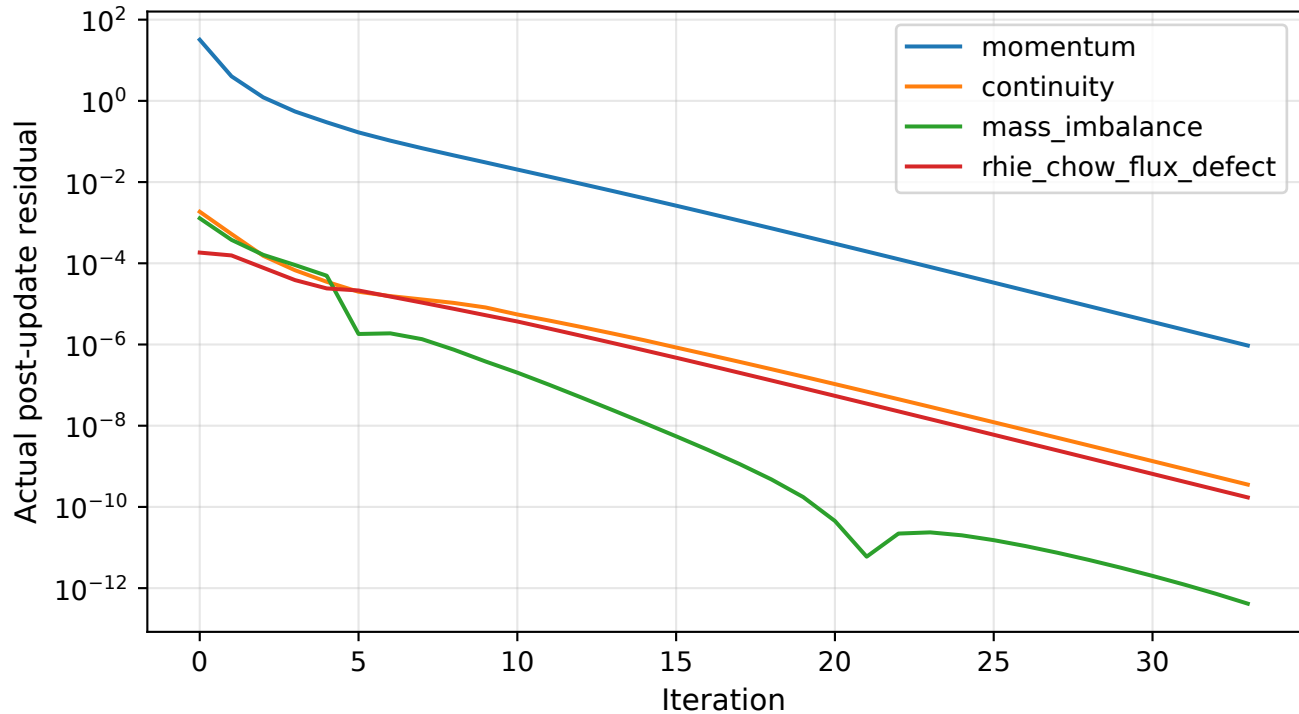
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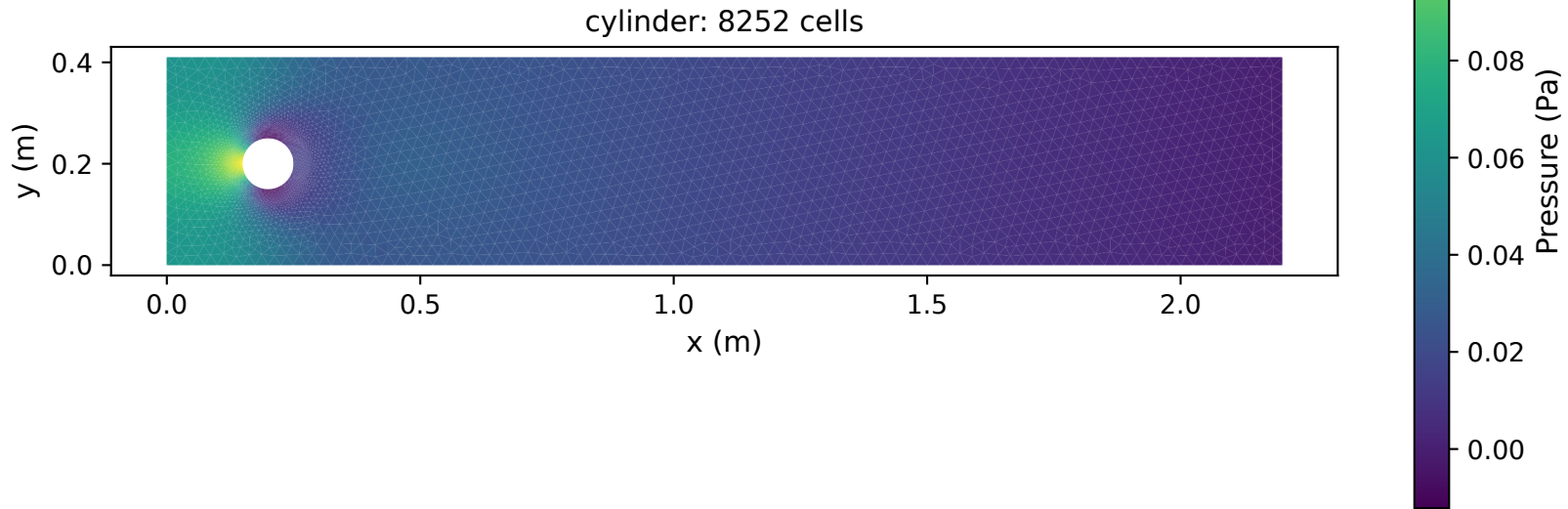


Actual body-fitted mesh

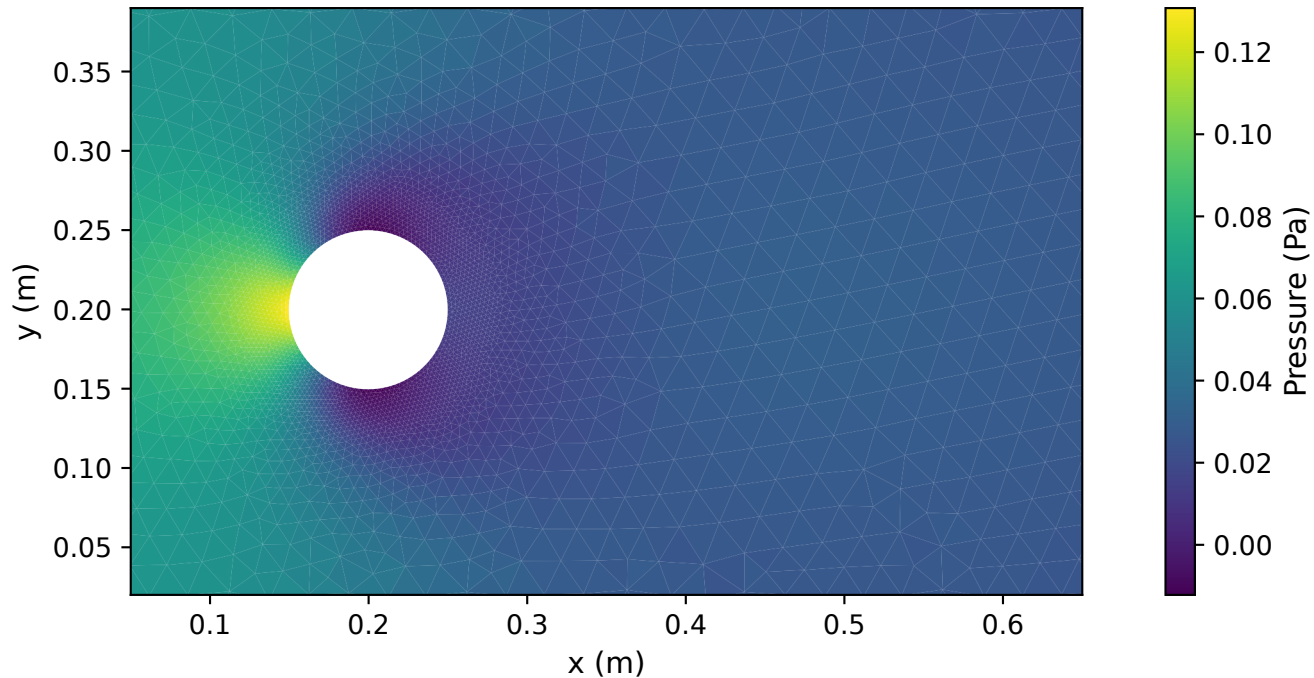


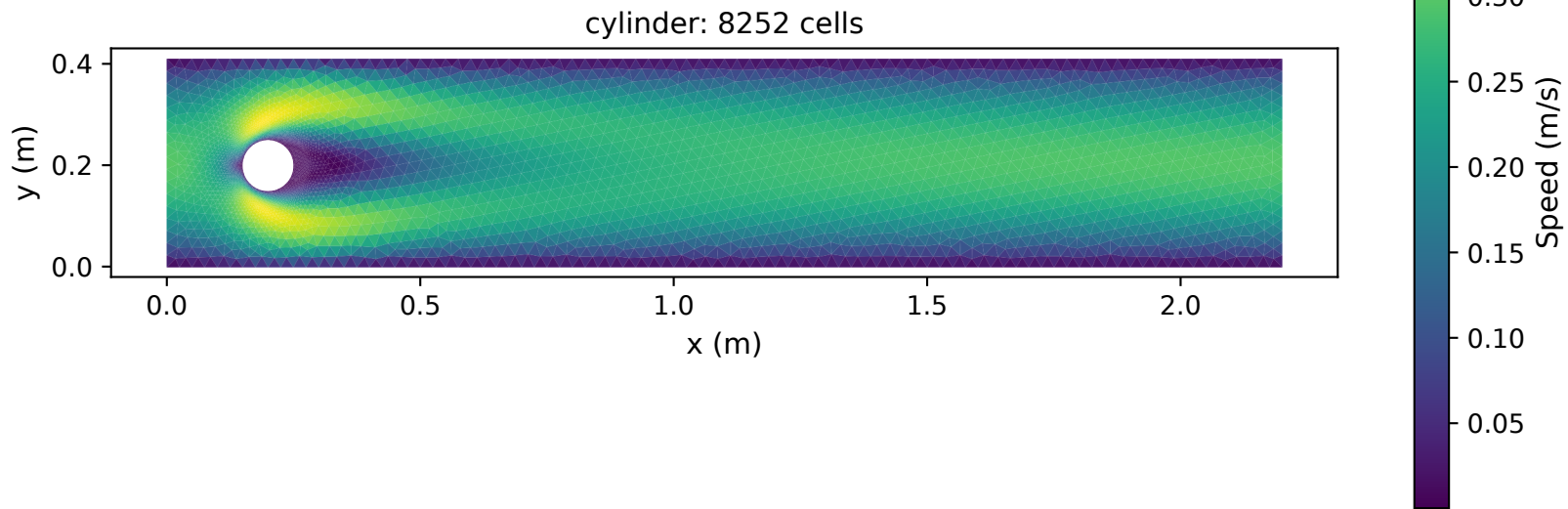




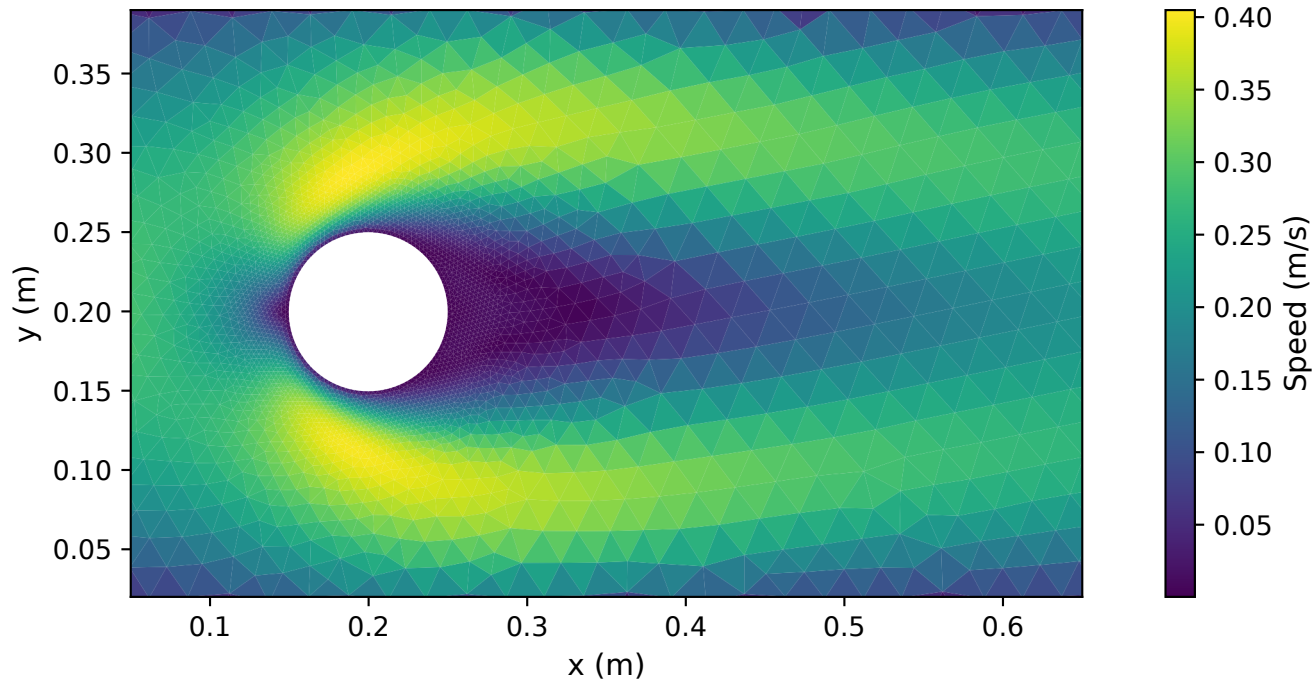


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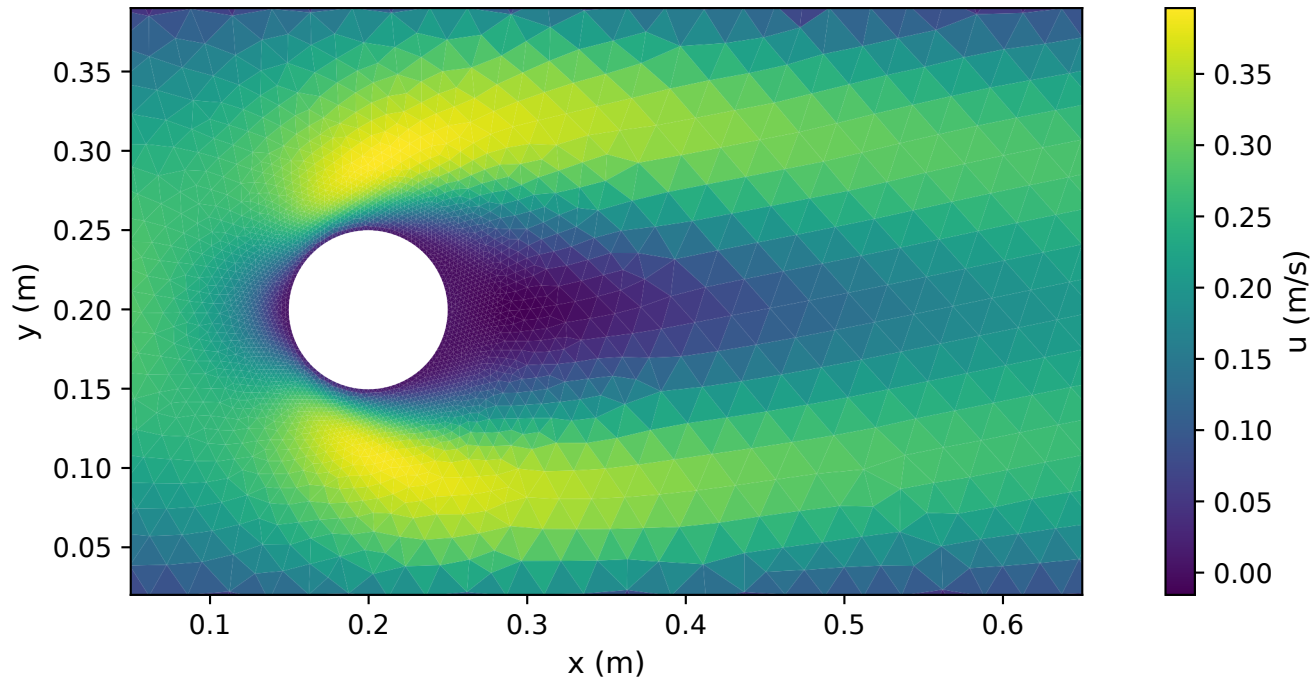




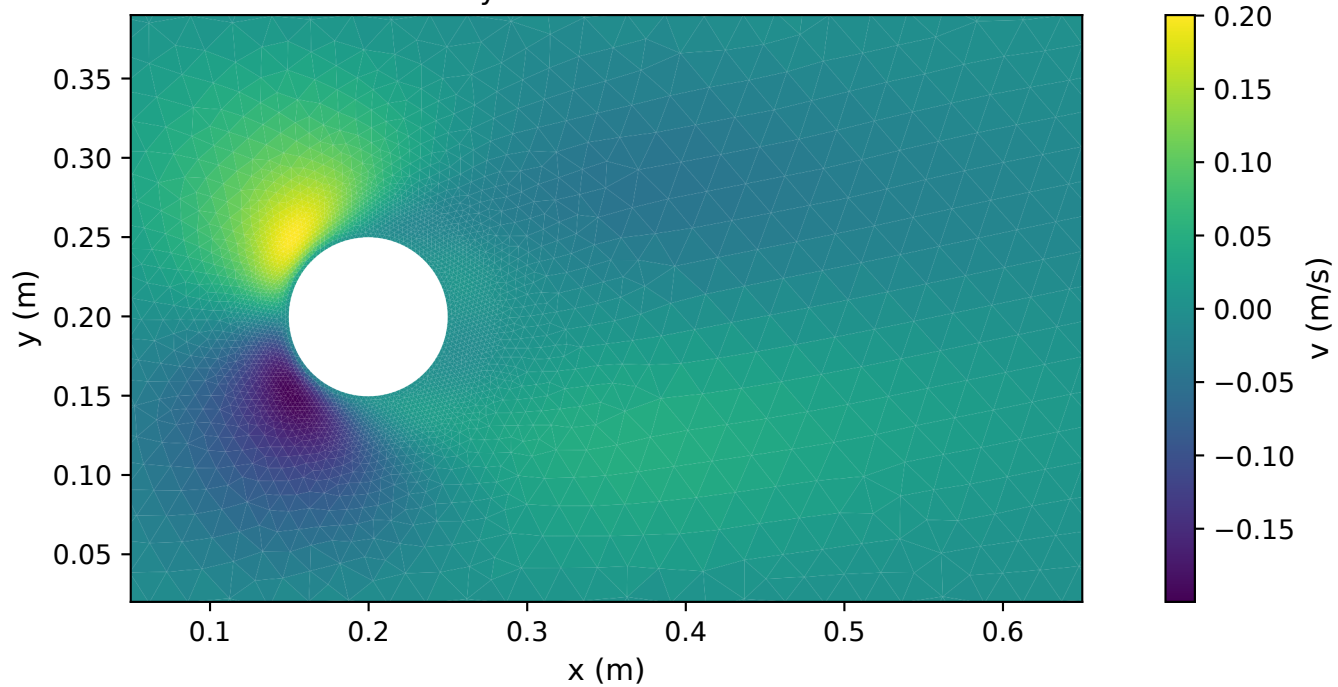
cylinder: 8252 cells



cylinder: 8252 cells



cylinder: 8252 cells



Actual body-fitted mesh

